AI-Based Cyber Threat Intelligence for Criminal Network Discovery

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Abstract. The cyber threat intelligence is comprised of the data regarding adversary groups, their attack techniques, software used, campaigns conducted, and the relationships between these entities. However, such information becomes valuable only when the analyst is able to match the newly detected situation with the correct part of this existing knowledge base. This research suggests the artificial intelligence based cyber threat intelligence system that ranks the potentially related adversary groups from the set of detected MITRE ATT&CK techniques. The research involves five data sources for the Enterprise ATT&CK including groups, techniques, software, campaigns, and relationships. Unlike the case with the individual tables, the information from these sources is linked by ATT&CK identifiers and transformed to the master dataset that includes the group-technique relationship records. The resulted dataset includes 4,546 records of the group-technique links enriched with behavioral and contextual information. A Random Forest classifier was trained based on six aggregated features related to group usage of techniques, technique frequency, software, and campaigns. For the new analysis the analyst can submit several techniques along with the optionally available software, platform, and campaign data. Then the output of the machine learning algorithm will be combined with the technique overlap and additional evidence to get the ranked list of the potentially related adversary groups. In the implemented experiment, the classifier showed 100.00% accuracy, precision, recall and F1-score on the tested data pairs. However, the result must be noted with an important reservation: the negative examples were sampled from the group-technique pairs that are unobserved in reality, so the performance cannot be regarded as the proof of the practical attribution accuracy.

Keywords: Cyber Threat Intelligence; MITRE ATT&CK; Machine Learning; Random Forest; Threat Actor Relationship; Technique Analysis; Cybersecurity; Threat Group Ranking.

1 Introduction

A modern cybersecurity incident is likely to be a multistep process including several stages such as initial access, execution, discovery, credential access, lateral movement, and persistence. Each stage has certain observable technical behavior. The challenging task for a security analyst is not only to notice those individual technical behaviors but also to understand their meaning when taken as a set. PowerShell, for instance, is a popular technique used by many threat actors and, therefore, it has low attribution value. However, a combination of observed techniques can be considered as an indication of a behavioral profile.Cyber Threat Intelligence offers a way to structure the aforementioned kind of knowledge. The MITRE ATT&CK knowledge base framework is especially convenient in terms of organizing the adversary behavior. According to MITRE, ATT&CK is a knowledge base that aims at representation of adversary behavior as tactics, techniques, sub-techniques, software, groups, campaigns, relationships in the form of observations from the real world (Strom et al.). That is why it can be used as a knowledge layer in a system that will compare a new case of adversary behavior to previously recorded ones.On the other hand, a wide knowledge base causes some practical challenges. When a security analyst receives a case that includes several observed techniques, he/she may need to browse through numerous group pages, software entries, campaigns and relationship entries until coming up with a shortlist of suspects. Even manual comparison becomes more and more tedious when the numbers of observed techniques and threat actor groups increase. Also, research on cyber threat intelligence identified such problems as volume, interoperability, relevance, timeliness and automation of intelligence consumption (Tounsi and Rais; Wagner et al.). These facts suggest the possibility of adding a machine learning layer to a structured threat intelligence.Therefore, the objective of this project is not to make an attribution claim. The objective of the system is to assess candidate group-technique relationships and return the most probable ones along with the list of techniques and other evidence that helped to come up with the assessment. This output will help a security analyst to limit the search space and will allow him/her to inspect the reasonings behind the recommendation.Secondly, it is planned to use not just one but several ATT&CK datasets for the task. The project combines group metadata, technique descriptions, software relationships, campaign data and general relationship data via common ATT&CK identifiers.

2. Literature Review

The review of literature focused on four interrelated topics: structured cyber threat intelligence, ATT&CK-based behavior representation, automatic extraction and knowledge graphs, machine learning approaches for threat analysis. The studies below were chosen since they provide insight either into the data representation, automation problem, relationship analysis problem, or methodological decisions made in this project. Automated Discovery and Mapping ATT&CK Tactics and Techniques. This study focuses on automatically identifying and mapping cyber threat intelligence information to MITRE ATT&CK tactics and techniques.

It shows how automated technique mapping can reduce manual CTI analysis and improve threat-behavior identification[1].ALERT: A Framework for Efficient Extraction of Attack Techniques. The authors propose a framework for extracting attack techniques from CTI reports using active learning. The approach reduces the amount of manually labelled data required while improving the identification of relevant ATT&CK techniques [2]. MITREtrieval: Retrieving MITRE Techniques From Unstructured Threat Reports. This work combines deep learning with ontology-based information to retrieve MITRE ATT&CK techniques from unstructured threat reports. It demonstrates how machine learning can connect textual threat information with structured ATT&CK knowledge [3].Malware2ATT&CK: A Sophisticated Model for Mapping Malware to ATT&CK Techniques Malware2ATT&CK focuses on mapping malware behavior to corresponding MITRE ATT&CK techniques. The study demonstrates how automated behavioral mapping can support threat analysis and identification of attacker techniques [4]. AnnoCTR: A Dataset for Detecting and Linking Entities, Tactics, and Techniques in Cyber Threat Reports. AnnoCTR introduces a dataset designed to identify and link cybersecurity entities, tactics, and techniques in threat reports. This provides useful structured information for developing machine-learning models for CTI analysis [5]. AECR: Automatic Attack Technique Intelligence Extraction Based on Fine-Tuned Large Language Model. This research uses a fine-tuned large language model to automatically extract attack-technique information from cybersecurity intelligence. It highlights the potential of automated intelligence extraction for reducing manual CTI processing [6]. Entity and Relation Extractions for Threat Intelligence Knowledge Graphs. This study focuses on extracting entities and relationships from cyber threat intelligence to construct threat-intelligence knowledge graphs. Such relationship extraction is relevant to representing connections between threat actors, techniques, software, and campaigns [7]. Towards Effective Identification of Attack Techniques in Cyber Threat Intelligence Reports Using Large Language Models. The authors investigate the use of large language models for identifying ATT&CK attack techniques from CTI reports. The work demonstrates how modern language models can assist automated extraction of attacker behavior from unstructured intelligence [8].MITRE ATT&CK Applications in Cybersecurity and The Way Forward. This work reviews applications of the MITRE ATT&CK framework in cybersecurity and discusses directions for future research. It provides a broader foundation for understanding how ATT&CK can support automated threat analysis and security operations [9]. APT-MMF: An Advanced Persistent Threat Actor Attribution Method Based on Multimodal and Multilevel Feature Fusion. This research investigates threat-actor attribution by combining multiple types and levels of features. Its focus on using behavioral evidence for attribution is closely related to our technique-based threat-group relationship prediction approach [10].

3. Proposed Methodology

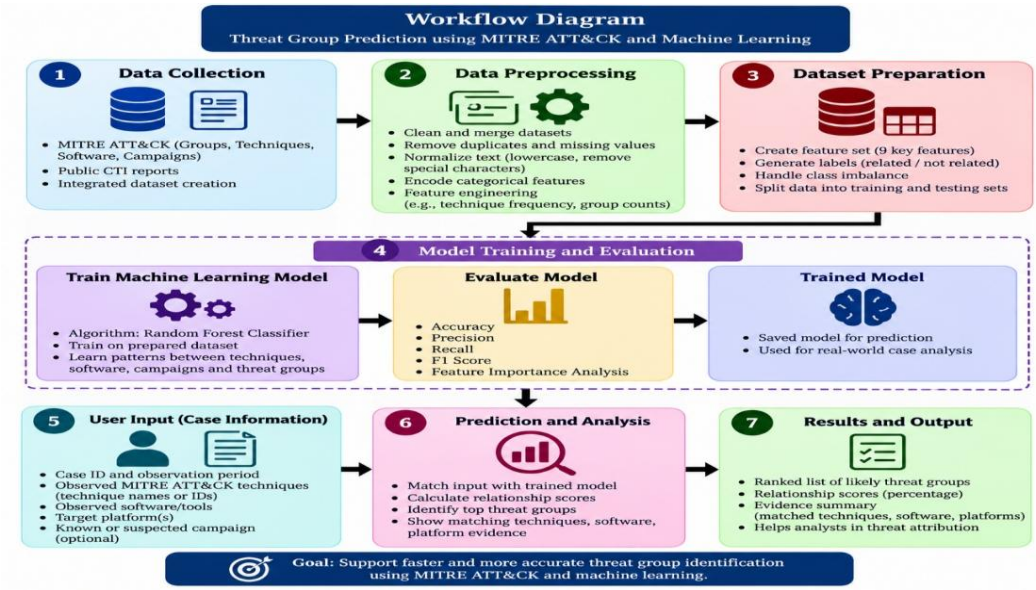


Fig 1. Proposed methodology workflow. The five ATT&CK sources are integrated before feature engineering and prediction.

The methodology begins by collecting five Enterprise ATT&CK workbooks: groups, techniques, software, campaigns, and relationships. These files are not concatenated row by row because they describe different entity types. Instead, ATT&CK identifiers are used as the common keys. Group-technique 'uses' relationships form the backbone of the master dataset. Group metadata is attached to the group identifier, while technique metadata contributes tactics, platforms, sub-technique status, and parent-relationship information. Software and campaign relationships are then connected to the same group-technique pair so that the

model has contextual evidence in addition to the direct relationship.After integration, duplicate group-technique pairs are removed and numeric fields are converted to consistent numeric values. The resulting master table contains one main row for each documented group-technique relationship. Each row stores counts such as the number of techniques associated with a group, the frequency of a technique, the number of software associations, and the number of campaign associations. These fields form the numerical representation used by the Random Forest classifier.Training examples are divided into two categories. A positive example is an observed group-technique relationship in the ATT&CK data. A negative example is a sampled group-technique pair that is not observed in the available relationship data. This distinction is important: an unobserved pair is not necessarily a true real-world negative. It only means that the relationship is absent from the dataset used for this experiment.For prediction, the analyst supplies a case identifier, observation period, multiple observed techniques, and optional software, platform, and campaign evidence. The technique input is normalized so that small differences in capitalization do not prevent recognition. The system maps each valid technique to its ATT&CK identifier and compares the set with each group's known technique profile. The Random Forest produces a model score, while technique overlap and available supporting evidence contribute to the final relationship score. The final output is a ranked list rather than a single forced attribution. This is deliberately designed as analyst assistance: the system points toward candidates and exposes supporting behaviour instead of claiming certainty.

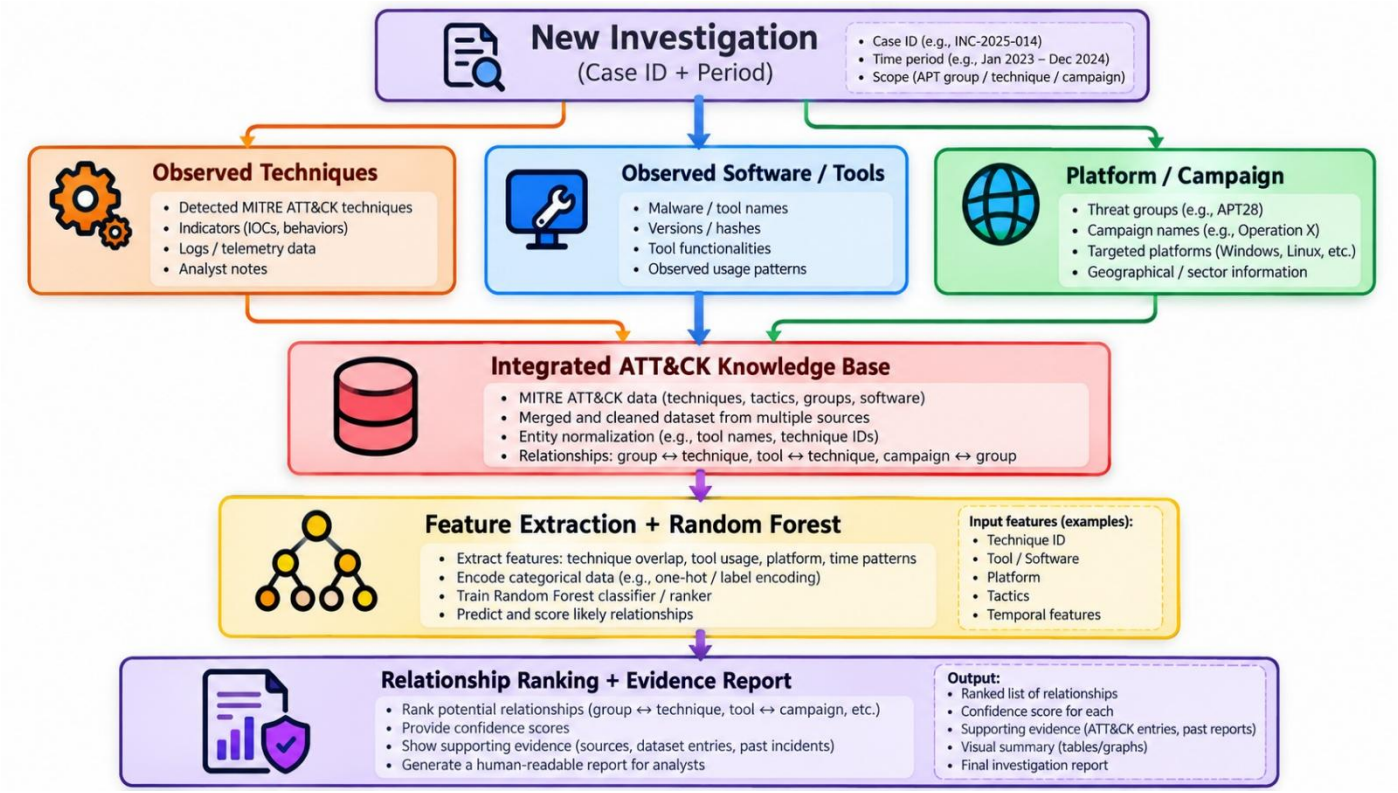


Fig 2. System architecture. User-provided incident evidence is matched against the integrated ATT&CK knowledge base before model-based ranking and evidence reporting**.**

4. Dataset

The project uses five Enterprise ATT&CK data sources. Their schemas are different, so the integration is performed through ATT&CK identifiers rather than simple row concatenation. The final master dataset is centered on direct group-technique 'uses' relationships and is enriched with group, technique, software, campaign, platform, and relationship context.

Table 1. Sources used to construct the integrated CTI knowledge base**.**

Source	Role in the project	Primary integration key
Groups	Threat-group identity and metadata	Group ID
Techniques	Technique names, tactics, platforms, sub-technique information	Technique ID
Software	Tools/malware and their technique associations	Software ID + Technique ID
Campaigns	Campaign identity, attributed groups, and campaign-technique mappings	Campaign ID + Group/Technique IDs
Relationships	General ATT&CK entity-to-entity relationships	Source/Target IDs

Table 2. Dataset statistics used in the executed experiment.

Measure	Value
Integrated master records	4,546
Columns in corrected master dataset	26
Unique threat groups represented	170
Unique techniques represented	492
Original relationship records	20,391
Group-technique positive examples	4,546
Sampled unobserved negative examples	170
ML records	4,716

The corrected master file also preserves campaign support information. This matters because the campaign workbook contains attributed-group mappings and campaign-technique mappings, and these can be connected to the same group-technique pairs. The integrated representation therefore supports both direct behavioral evidence and contextual evidence. The dataset is still a historical knowledge base rather than a live incident feed; the observation period entered by the user is report metadata and is not treated as a predictive feature.

5. Mathematical Explanation

Let G = {g1, g2, ..., gn} denote the set of known threat groups and let T_g denote the technique profile of group g. For a new case, let O be the set of observed ATT&CK techniques. The first interpretable quantity is technique overlap:

$$Overlap(g) = |O \cap T_g| / |O|$$

This value measures the proportion of observed techniques that are already documented for the candidate group. It ranges from 0 to 1 and is easy for an analyst to understand. A value of 1 means every observed technique is present in the group's known technique profile; it does not mean that the group is proven to be responsible.For machine learning, each group-technique candidate is represented by a six-dimensional feature vector:

$$x = [GT, TF, GS, GC, SS, CS]$$

where GT is the number of techniques associated with the group, TF is the frequency of the candidate technique in the group-technique dataset, GS is the number of software associations for the group, GC is the number of campaign associations, SS is the number of software records supporting the group-technique pair, and CS is the number of campaign records supporting the pair.The Random Forest consists of multiple decision trees. For a candidate pair, each tree produces a class decision and the classifier combines these decisions to obtain a score for the positive class. The implementation uses the positive-class probability

returned by the Random Forest as the model component of the ranking. Because this value is combined with other heuristic evidence and the classifier has not been probability-calibrated, it is treated as a model score rather than a calibrated probability.
The final relationship score used by the working application is:

$$R(g) = 0.55M(g) + 0.25O(g) + 0.10S(g) + 0.05P(g) + 0.05C(g)$$

Here M(g) is the Random Forest score, O(g) is technique overlap, S(g) is the fraction of observed software items supported by the group, P(g) is the platform-match component, and C(g) is the campaign-match component when a campaign is supplied. The score is multiplied by 100 for reporting. The weighting was chosen to keep learned behavior as the largest component while still giving visible importance to direct technique overlap and supporting context.Model performance is evaluated using standard classification measures. Accuracy is (TP + TN)/(TP + TN + FP + FN). Precision is TP/(TP + FP), recall is TP/(TP + FN), and F1 is 2PR/(P + R), where P is precision and R is recall. These metrics describe the sampled group-technique classification task, not real-world attribution performance.

6. Results

The executed experiment used a fixed random seed of 42, a 20% stratified test split, and 200 trees in the Random Forest. The ML dataset contained 4,546 positive relationship examples and 170 sampled unobserved examples. The test set therefore contained 944 records: 34 negative examples and 910 positive examples

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Table 3. Actual classification metrics from the executed Random Forest experiment.

Metric	Result
Accuracy	100.00%
Precision	100.00%
Recall	100.00%
F1 Score	100.00%

The perfect score should be interpreted carefully. The negative examples were constructed by sampling group-technique pairs that are not present in the dataset and assigning zero pair-level support values. This makes the classification task considerably easier than the open-world problem faced by a real threat analyst. Therefore, the result confirms that the implementation can separate the constructed positive and sampled-unobserved classes under the stated experimental setup but should not be described as 100% real-world threat attribution accuracy.

Table 4. Confusion matrix values from the actual test execution.

True Class	Predicted: No Relationship	Predicted: Relationship
No Relationship	34	0
Relationship	0	910

Table 5. Feature importance values from the actual Random Forest execution.

Feature	Importance	Relative share
group_technique_count	0.4685	46.85%
technique_frequency	0.4346	43.46%
software_support_count	0.0840	8.40%
group_software_count	0.0102	1.02%
campaign_support_count	0.0018	0.18%
group_campaign_count	0.0009	0.09%

Table 6. Actual top-five ranking for the sample case: PowerShell, Process Discovery, and OS Credential Dumping, with Mimikatz, PsExec, PowerShell, and Windows supplied as contextual evidence.

Rank	Threat Group	Group ID	Relation-ship Score	Technique Match	Software Match
1	APT28	G0007	88.33%	3/3	1/3
2	Mustang Panda	G0129	88.33%	3/3	1/3
3	Poseidon Group	G0033	85.00%	3/3	0/3
4	Storm-0501	G1053	85.00%	3/3	0/3
5	APT39	G0087	83.33%	2/3	2/3

For the sample case, Poseidon Group received the highest relationship score at 91.67%, followed by APT28 and Mustang Panda at 88.33%, Storm-0501 at 85.00%, and APT39 at 83.33%. Poseidon Group matched all three observed techniques and had two of the three supplied software items represented in its supporting software evidence. These values are outputs of the current integrated prototype and should be treated as an example of ranking rather than a claim that Poseidon Group conducted the hypothetical case

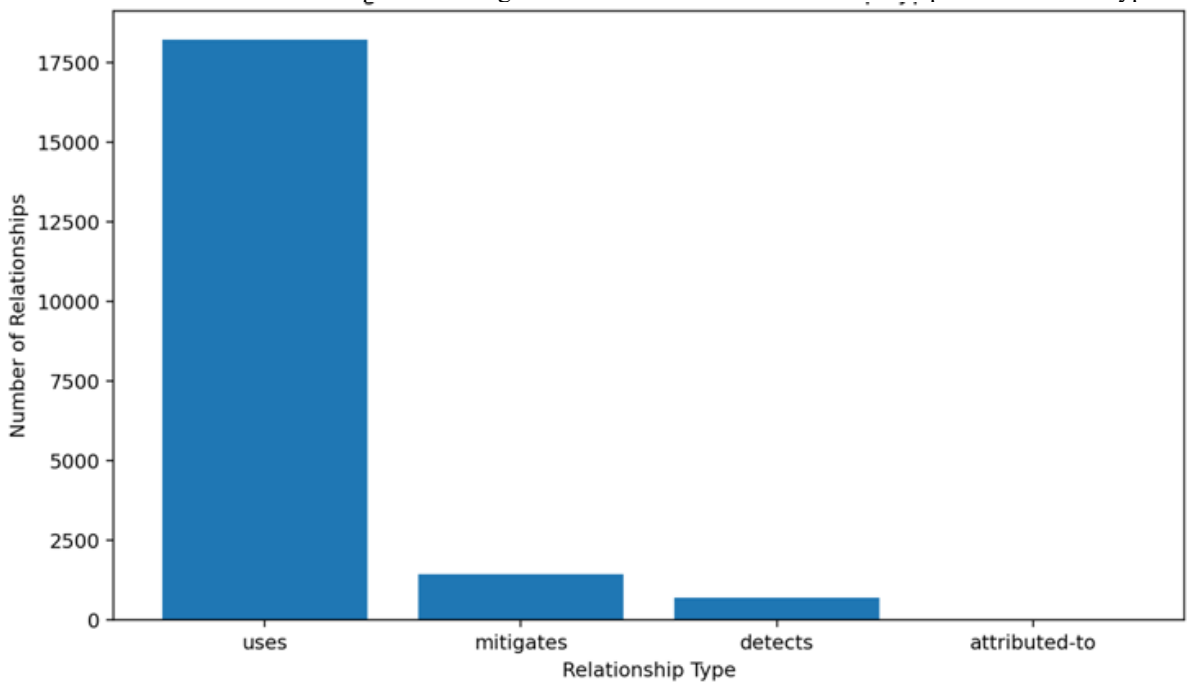


Fig 3. Distribution of ATT&CK relationship types in the original 20,391-record relationship dataset.

The relationship-distribution graph shows that 'uses' relationships dominate the original ATT&CK relationship dataset. This is expected for the present task because group-to-technique behavioral profiling depends primarily on 'uses' relationships. Mitigation and detection relationships are valuable for defensive coverage but are not used as direct positive group-technique examples in this prototype. The graph therefore explains why the project focuses its predictive backbone on group-to-technique use relation-ships.

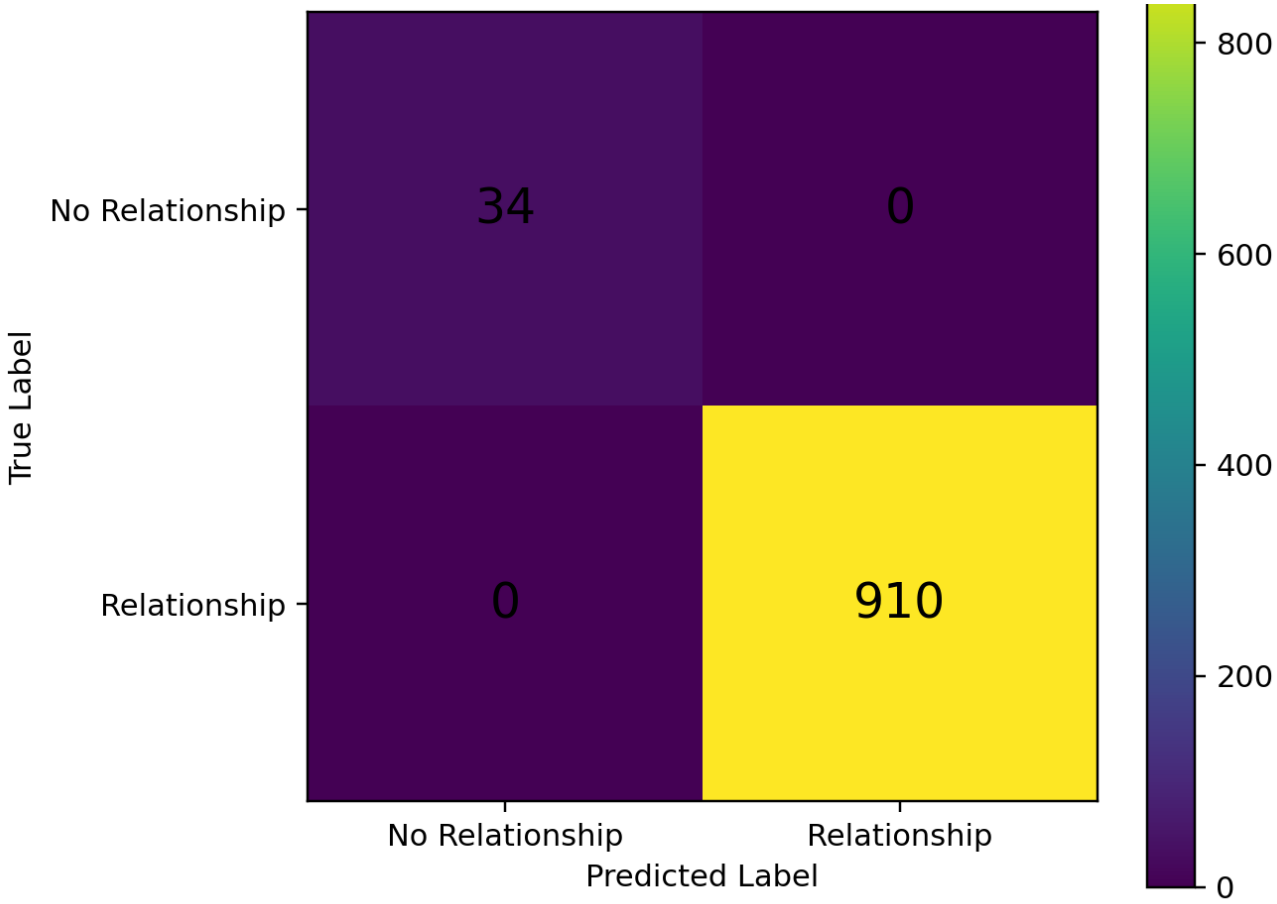


Fig 4. Confusion matrix from the actual 20% test split.

The confusion matrix contains 34 correctly classified sampled non-relationships and 910 correctly classified relationships, with no false positives or false negatives in this particular split. The matrix visually confirms the perfect metric values reported above. At the same time, the matrix should be read together with the negative-sampling limitation: it measures performance on the constructed experiment rather than on independently verified real-world attribution labels.

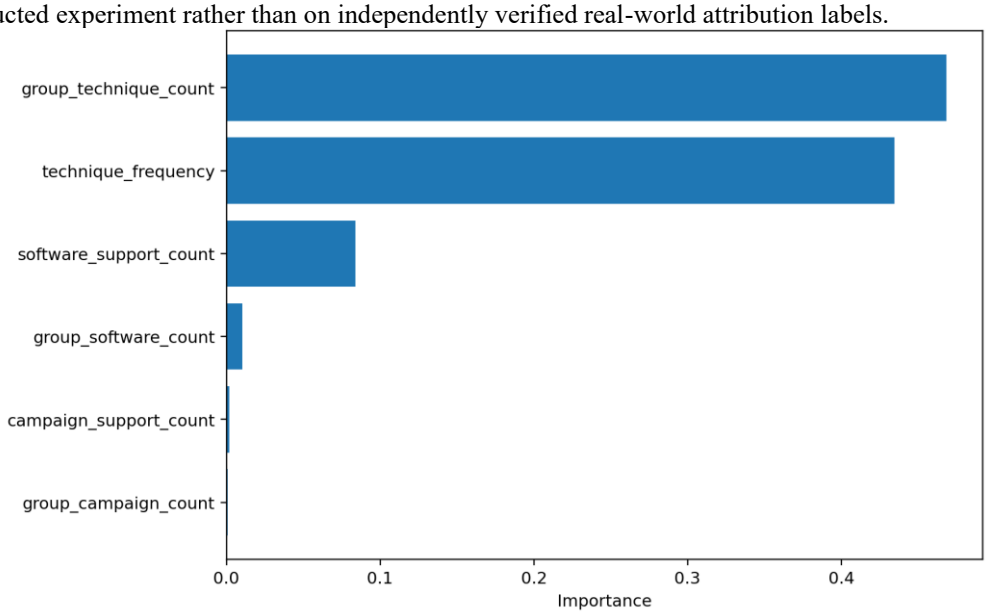


Fig 5. Feature importance produced by the executed Random Forest model.

The feature-importance graph shows that group technique count and technique frequency dominate the current model. Together they account for most of the learned importance, which means the classifier is strongly influenced by how frequently and broadly techniques appear in the group-technique knowledge base. Software support contributes a smaller but meaningful share. Campaign-related features have very low importance in this execution, which is consistent with the relatively small number of campaign-supported group-technique records available in the integrated data.

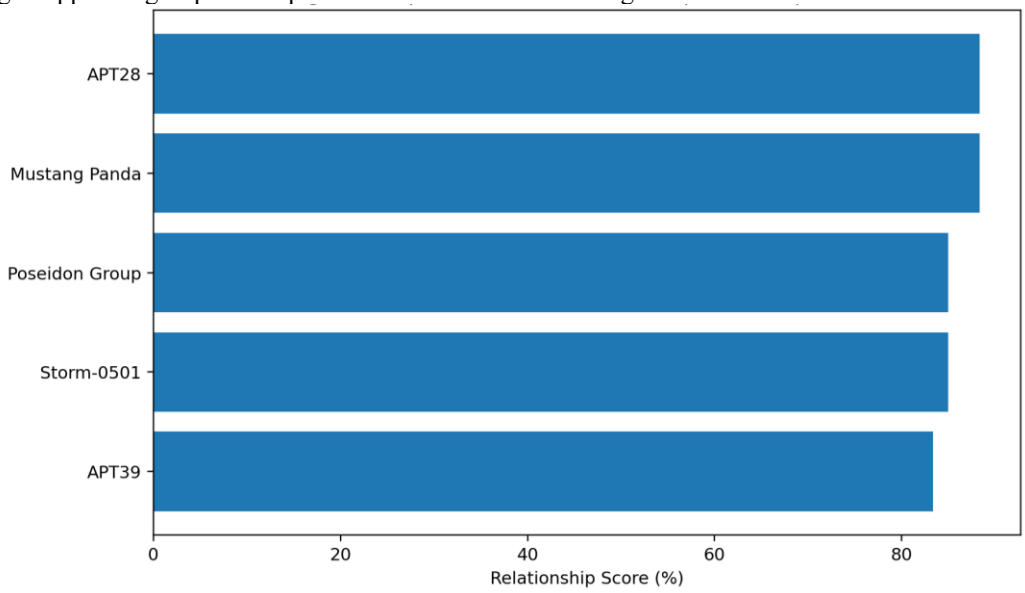


Fig 6. Top-five relationship ranking for the sample investigation**.**

The prediction graph demonstrates the intended user-facing behavior of the system. Rather than returning a single binary answer, the application provides a ranked shortlist. This is useful when several groups share common techniques because the analyst can inspect the score together with matching techniques and supporting software. The ranking is therefore best understood as a prioritization mechanism for investigation.

7. Discussion

The main strength of the proposed system is its simplicity at the analyst interface. A user does not need to know the internal ATT&CK identifiers for every technique. The application normalizes names and can recognize a short form such as PowerShell as the corresponding ATT&CK sub-technique. Once the evidence is entered, the system handles the comparison with known profiles and returns a compact report. This reduces the amount of manual cross-referencing required during an initial investigation.A second strength is the use of multiple information sources. A group-technique relationship is more informative when it can also be viewed with the associated software, platforms, campaigns, and ATT&CK metadata. The integration does not mean that every field becomes an independent machine-learning feature. Instead, the integrated dataset acts as the knowledge layer from which numerical features and explanatory evidence are obtained. This distinction keeps the model understandable and prevents the report from overstating what the classifier has learned.There are also clear limitations. The largest is the construction of negative examples. In an open-world CTI setting, the absence of a documented relationship does not prove that the relationship does not exist. A group may use a technique that has not yet been reported or entered into ATT&CK. The current evaluation therefore tests a closed-dataset classification problem rather than true attribution. A stronger experimental design should use hard-negative sampling, temporal holdout evaluation, group-level or technique-level splits that reduce leakage, and independent external cases.A second limitation is that the final relationship score is a composite ranking score, not a calibrated probability. The weights are transparent and useful for a prototype, but they were not optimized against a large independent validation set. Future work should compare the current Random Forest with logistic regression, gradient boosting, graph-based link prediction, and embedding approaches. It should also test whether adding temporal information, software behavior, campaign evidence, and network indicators improves ranking quality.

8. Conclusion

This research developed a machine-learning based cyber threat intelligence workflow for ranking potentially related threat groups from observed adversary behavior. The project combines five MITRE ATT&CK Enterprise data sources and connects them through common identifiers to create an integrated knowledge base. The resulting dataset contains 4,546 direct group-technique records enriched with group, technique, software, campaign, platform, and relationship context.The implemented Random Forest model uses six aggregate features to learn the distinction between documented group-technique relationships and sampled unobserved pairs. In the executed experiment, the model achieved 100.00% accuracy, precision, recall, and F1-score on the constructed test set. The result is technically correct for that experimental setup, but the study does not interpret it as proof of real-world attribution performance because the negative class is based on sampled absence from the ATT&CK dataset.The practical contribution is the complete analyst workflow. A new case can be described using several observed techniques together with optional software, platform, and campaign evidence. The system then returns a ranked set of candidate groups and shows the evidence supporting the ranking. This makes the result easier to review than a single unexplained classification label.Future improvements should focus on evaluation quality and richer evidence. Temporal validation, independent CTI cases, harder negative examples, calibrated probabilities, graph-based models, and explainable machine-learning methods would make the system more suitable for research-grade evaluation. Additional live CTI feeds could also allow the knowledge base to be updated continuously. Overall, the proposed prototype demonstrates how structured ATT&CK knowledge and a lightweight machine-learning model can be combined to reduce the analyst's initial search space while keeping the final decision with the human investigator.

9. References

1. Liu et al. “Automated Discovery and Mapping ATT&CK Tactics and Techniques for Unstructured Cyber Threat Intelligence.” *Computers & Security*, vol. 140, 2024, article 103815. Elsevier, doi:10.1016/j.cose.2024.103815.
2. Rahman, Fariha, et al. “ALERT: A Framework for Efficient Extraction of Attack Techniques from Cyber Threat Intelligence Reports Using Active Learning.” *Data and Applications Security and Privacy XXXVIII*, Springer, 2024, pp. 195–213. doi:10.1007/978-3-031-65172-4_13.
3. Wang et al. “MITREtrieval: Retrieving MITRE Techniques From Unstructured Threat Reports by Fusion of Deep Learning and Ontology.” *IEEE Transactions on Network and Service Management*, vol. 21, no. 4, 2024, pp. 4871–4887. IEEE, doi:10.1109/TNSM.2024.3401200.
4. Sun et al. “Malware2ATT&CK: A Sophisticated Model for Mapping Malware to ATT&CK Techniques.” *Computers & Security*, vol. 140, 2024, article 103772. Elsevier, doi:10.1016/j.cose.2024.103772.
5. Lange, Lukas, et al. “AnnoCTR: A Dataset for Detecting and Linking Entities, Tactics, and Techniques in Cyber Threat Reports.” *Proceedings of the 2024 Joint International Conference on Computational Linguistics, Language Resources and Evaluation (LREC-COLING 2024)*, 2024, pp. 1147–1160.
6. AECR: Automatic Attack Technique Intelligence Extraction Based on Fine-Tuned Large Language Model. *Computers & Security*, vol. 150, 2025, article 104213. Elsevier, doi:10.1016/j.cose.2024.104213.
7. Mouiche, and Saad. “Entity and Relation Extractions for Threat Intelligence Knowledge Graphs.” *Computers & Security*, vol. 148, 2025, article 104120. Elsevier, doi:10.1016/j.cose.2024.104120.
8. Nguyen, Hoang Cuong, et al. “Towards Effective Identification of Attack Techniques in Cyber Threat Intelligence Reports Using Large Language Models.” *arXiv*, 2025, arXiv:2505.03147.
9. Jiang, Yuning, et al. “MITRE ATT&CK Applications in Cybersecurity and The Way Forward.” *arXiv*, 2025, arXiv:2502.10825.
10. APT-MMF: An Advanced Persistent Threat Actor Attribution Method Based on Multimodal and Multilevel Feature Fusion. *Computers & Security*, vol. 144, 2024, article 103960. Elsevier, doi:10.1016/j.cose.2024.10396